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Aim: Configure IPsec on network devices to provide secure communication and protect against unauthorized access and attacks.

Code: CLI

```
import random

def generate_private_key():
    return random.randint(2, 100)

def generate_public_key(private_key, g, p):
    return pow(g, private_key, p)

def generate_shared_key(public_key, private_key, p):
    return pow(public_key, private_key, p)

def main():
    print("DIFFIE-HELLMAN KEY EXCHANGE")
    print("=" * 40)

    p = int(input("Enter prime number p: "))
    g = int(input("Enter primitive root g: "))

    alice_private = generate_private_key()
    bob_private = generate_private_key()

    alice_public = generate_public_key(
        alice_private,
        g,
        p
    )
    bob_public = generate_public_key(
        bob_private,
        g,
        p
    )

    alice_shared = generate_shared_key(
        bob_public,
        alice_private,
```

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```
p
)

bob_shared = generate_shared_key(
    alice_public,
    bob_private,
    p
)

print("\nPurvi Private Key:", alice_private)
print("Purvi Public Key:", alice_public)

print("\nNeha Private Key:", bob_private)
print("Neha Public Key:", bob_public)

print("\nPurvi Shared Key:", alice_shared)
print("Neha Shared Key:", bob_shared)

print("\nResult:")

if alice_shared == bob_shared:
    print("Key exchange successful")
    print("Both parties have the same shared secret.")
else:
    print("Key exchange failed")

if __name__ == "__main__":
    main()
```

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Output:

```
IDLE Shell 3.12.4
File Edit Shell Debug Options Window Help
Python 3.12.4 (tags/v3.12.4:8e8a4ba, Jun 6 2024, 19:30:16) [MSC v.1940 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:\Users\Purvi\Downloads\prac5-T086 cli.py
DIFFIE-HELLMAN KEY EXCHANGE
=====
Enter prime number p: 23
Enter primitive root g: 3

Purvi Private Key: 49
Purvi Public Key: 13

Neha Private Key: 87
Neha Public Key: 8

Purvi Shared Key: 16
Neha Shared Key: 16

Result:
Key exchange successful
Both parties have the same shared secret.
>>>
```

```
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Key exchange successful
Both parties have the same shared secret.
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```

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Code: GUI

```
import tkinter as tk
from tkinter import messagebox
import random

def private_key():
    return random.randint(2, 10)

def public_key(private, g, p):
    return pow(g, private, p)

def shared_key(public, private, p):
    return pow(public, private, p)

def exchange():
    try:
        p = int(p_entry.get())
        g = int(g_entry.get())

        private1 = private_key()
        private2 = private_key()

        public1 = public_key(private1, g, p)
        public2 = public_key(private2, g, p)

        shared1 = shared_key(public2, private1, p)
        shared2 = shared_key(public1, private2, p)

        private1_entry.delete(0, tk.END)
        private1_entry.insert(0, private1)

        public1_entry.delete(0, tk.END)
        public1_entry.insert(0, public1)

        private2_entry.delete(0, tk.END)
        private2_entry.insert(0, private2)

        public2_entry.delete(0, tk.END)
        public2_entry.insert(0, public2)

        shared1_entry.delete(0, tk.END)
        shared1_entry.insert(0, shared1)
```

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```
shared2_entry.delete(0, tk.END)
shared2_entry.insert(0, shared2)
```

```
if shared1 == shared2:
    result.config(
        text="KEY EXCHANGE SUCCESSFUL\nPurvi and Neha have the same secret key.",
        fg="green"
    )
else:
    result.config(
        text="KEY EXCHANGE FAILED",
        fg="red"
    )
```

```
except ValueError:
    messagebox.showerror("Error", "Please enter valid numbers")
```

```
root = tk.Tk()
root.title("Diffie-Hellman Key Exchange")
root.geometry("600x650")
root.config(bg="lightgray")
```

```
tk.Label(
    root,
    text="DIFFIE-HELLMAN KEY EXCHANGE",
    font=("Arial", 20, "bold"),
    bg="lightgray"
).pack(pady=15)
```

```
tk.Label(
    root,
    text="Prime Number (p)",
    bg="lightgray"
).pack()
```

```
p_entry = tk.Entry(root)
p_entry.pack()
p_entry.insert(0, "23")
```

```
tk.Label(
    root,
    text="Primitive Root (g)",
```

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```
        bg="lightgray"
    ).pack()

    g_entry = tk.Entry(root)
    g_entry.pack()
    g_entry.insert(0, "5")

    person1_label = tk.Label(
        root,
        text="PURVI",
        font=("Arial", 14, "bold"),
        bg="lightgray"
    )
    person1_label.pack(pady=10)

    private1_entry = tk.Entry(root, width=40)
    private1_entry.pack()

    tk.Label(
        root,
        text="Purvi Private Key",
        bg="lightgray"
    ).pack()

    public1_entry = tk.Entry(root, width=40)
    public1_entry.pack()

    tk.Label(
        root,
        text="Purvi Public Key",
        bg="lightgray"
    ).pack()

    person2_label = tk.Label(
        root,
        text="NEHA",
        font=("Arial", 14, "bold"),
        bg="lightgray"
    )
    person2_label.pack(pady=10)

    private2_entry = tk.Entry(root, width=40)
    private2_entry.pack()
```

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```
tk.Label(  
    root,  
    text="Neha Private Key",  
    bg="lightgray"  
).pack()
```

```
public2_entry = tk.Entry(root, width=40)  
public2_entry.pack()
```

```
tk.Label(  
    root,  
    text="Neha Public Key",  
    bg="lightgray"  
).pack()
```

```
shared1_entry = tk.Entry(root, width=40)  
shared1_entry.pack(pady=5)
```

```
tk.Label(  
    root,  
    text="Purvi Shared Key",  
    bg="lightgray"  
).pack()
```

```
shared2_entry = tk.Entry(root, width=40)  
shared2_entry.pack(pady=5)
```

```
tk.Label(  
    root,  
    text="Neha Shared Key",  
    bg="lightgray"  
).pack()
```

```
tk.Button(  
    root,  
    text="Perform Key Exchange",  
    command=exchange,  
    bg="blue",  
    fg="white",  
    font=("Arial", 11, "bold")  
).pack(pady=15)
```

```
result = tk.Label(  
    root,
```

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```
text="Click the button to perform key exchange",  
font=("Arial", 11, "bold"),  
bg="lightgray",  
fg="blue"  
)  
result.pack()  
root.mainloop()
```

Output:

Diffie-Hellman Key Exchange

DIFFIE-HELLMAN KEY EXCHANGE

Prime Number (p)
23

Primitive Root (g)
5

PURVI

4
Purvi Private Key
4
Purvi Public Key

NEHA

4
Neha Private Key
4
Neha Public Key
3
Purvi Shared Key
3
Neha Shared Key

Perform Key Exchange

KEY EXCHANGE SUCCESSFUL
Purvi and Neha have the same secret key.